Koneru Lakshmaiah Education Foundation
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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	Ⅰ-Ⅲ Semester(2025-26)
Faculty Name	Ms. Alluri Swathi
Student Name	Sivarapu Rishil
Roll Number	2520090174
Branch	CSIT
Course Outcome	3

Case Study Topic: UrbanAI – Intelligent Smart City Infrastructure Optimization Platform using BFS

Work Done Summary:
Implemented Breadth First Search (BFS) in Java to model and analyze smart city transportation and road networks. Represented city locations and road connections using graph data structures.
Implemented BFS traversal to analyze connectivity between different city zones and identify reachable infrastructure routes efficiently. The system monitored road connectivity and transportation access across multiple smart city regions.
Compared graph-based BFS traversal with traditional searching approaches and observed improved efficiency in smart city route analysis and urban transportation management systems.

Implementation Details :

- Created graph structures using adjacency lists to represent city road networks.
- AddEdge() connects city zones and transportation routes in the graph.
- BFS() performs connectivity analysis between different smart city regions.
- Queue data structure is used for efficient BFS traversal operations.
- ConnectivityAnalysis() monitors reachable transportation routes
- Display methods generate smart city road network reports.
- The main method demonstrated graph creation, edge insertion, BFS traversal, and smart city connectivity analytics operations.

Result : Screenshots/Output:

```
====== SMART CITY ROAD NETWORK ======

City Zone 0 Connected To : 1 2
City Zone 1 Connected To : 0 3
City Zone 2 Connected To : 0 4
City Zone 3 Connected To : 1 5
City Zone 4 Connected To : 2
City Zone 5 Connected To : 3

====== BFS CONNECTIVITY ANALYSIS ======

Visited City Zone : 0
Visited City Zone : 1
Visited City Zone : 2
Visited City Zone : 3
Visited City Zone : 4
Visited City Zone : 5

====== SMART CITY CONNECTIVITY ANALYSIS COMPLETED ======
```

Time Complexity:

- Graph Edge Insertion Operation: O(1)
- Breadth First Search (BFS): O(V + E)
- Connectivity Analysis Operation: O(V + E)
- Adjacency List Representation provides efficient storage and faster traversal for large-scale smart city transportation networks.

GitHub Link : https://github.com/rishill10/DSA2_CO3

Student Signature	Faculty Signature